

CHAPTER 30

Communicable disease

Communicable or infectious diseases (also called transmissible diseases) include diseases resulting from the invasion of, and multiplication in, the body of any foreign organisms. Such foreign, disease-causing organisms are called **pathogens**. Some communicable diseases are said to be **contagious**; this means they are passed by direct contact with a person suffering from the disease, or by contact with something touched by the person. Other communicable diseases may be spread from person to person by **vectors**, intermediate hosts of the pathogen, such as mosquitoes or fleas.

Pathogens

Bacteria

The great majority of bacteria are harmless to humans; they are non-pathogenic. Indeed, many bacteria are essential to life on earth through their

role in decomposition of organic material and the cycling of the elements. Huge numbers of bacteria live on our skin, in our alimentary canals, and in other parts of the body. In the armpit of an adult male there are more than two million bacteria per square centimetre of skin surface, and in the intestine bacteria are so numerous that the major part of the contents consist of bacteria. These bacteria have no ill effect on our health, yet there are others that may cause illness or death when present in relatively small numbers.

Bacteria are all unicellular and can only be seen with a microscope. Under the light microscope, about all that can be seen of bacteria is the shape of their cells. Cell shape is used to classify bacteria into the following groups:

- **Cocci** (sing. coccus) have a cell with a spherical shape. The cells may occur singly, in pairs (diplococci), in clusters (staphylococci) or in chains (streptococci).

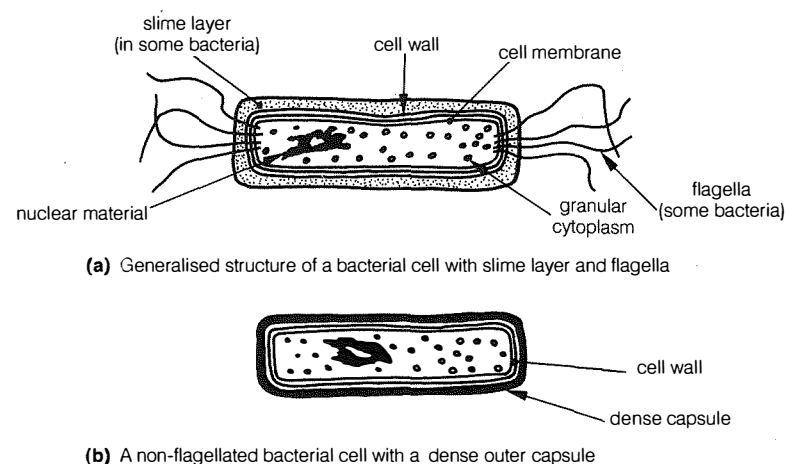


Fig. 30.1 Bacterial cell structure

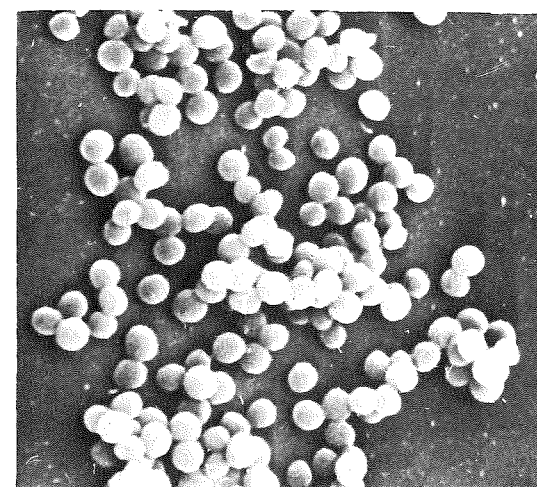


Fig. 30.2 *Staphylococcus aureus* (multiplication $\times 8500$), photographed using a scanning electron microscope. These spherical shaped bacterial cells occur in clusters in the upper respiratory tract of many people. Most strains do not cause disease but serious infections can occur when the resistance of the host person is low
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- **Bacilli** (sing. bacillus) are rod-shaped bacterial cells. Many of them have flagella and are **motile** (able to move).
- **Spirochaetes** are very fine filaments highly coiled into a tight spiral.
- **Spirilla** (sing. spirillum) are also twisted but not as tightly as the spirochaetes.

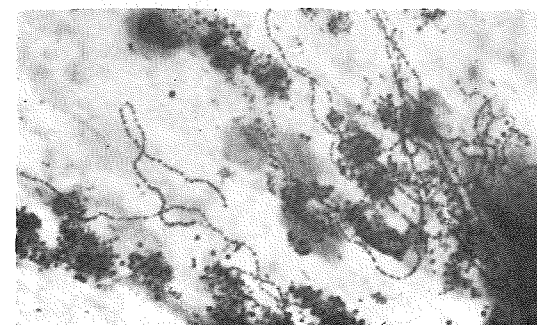


Fig. 30.3 Streptococci are spherical bacterial cells joined together in chains. Different species of streptococcus cause different diseases, such as sore throats and scarlet fever
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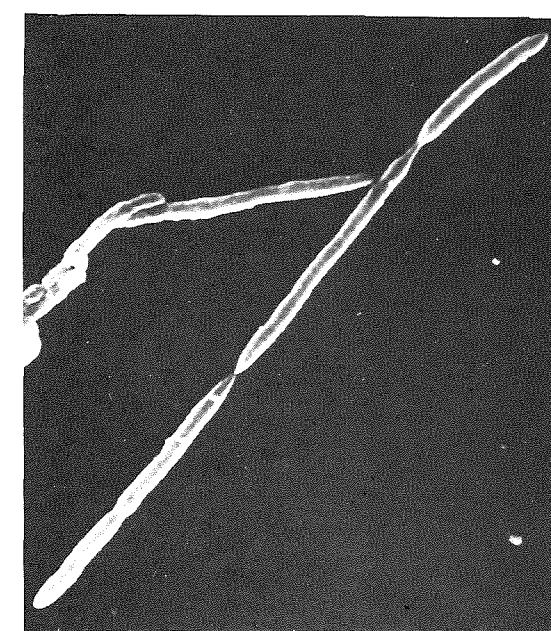


Fig. 30.4 Cells of a species of *Bacillus* joined together in chains ($\times 8500$)
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- **Vibrios** are less twisted again, being more like curved rods and often assuming a comma shape.

Structures inside a bacterial cell can only be seen with an electron microscope. The cell has a rigid cell wall, which is often covered on the outside by a slime layer. Within the cell wall the cytoplasm is surrounded by a cell membrane. Bacteria lack a true nucleus; there is nuclear material made of DNA (deoxyribose nucleic acid) in the cytoplasm, but it is not bounded by a nuclear membrane (see Fig. 30.7). Some bacteria are able to form spores by developing a resistant coat around the cell. Spore formation enables the cell to survive adverse conditions such as temperature extremes or desiccation. Bacterial cells reproduce by simple fission and under ideal conditions huge numbers can be produced in hours.

Compared with the great numbers of different species of bacteria, relatively few are pathogenic in humans. Pathogenic bacteria produce disease in a number of ways. In many cases the disease is the result of **toxins**—substances produced by the bacteria. Tetanus, diphtheria and bubonic

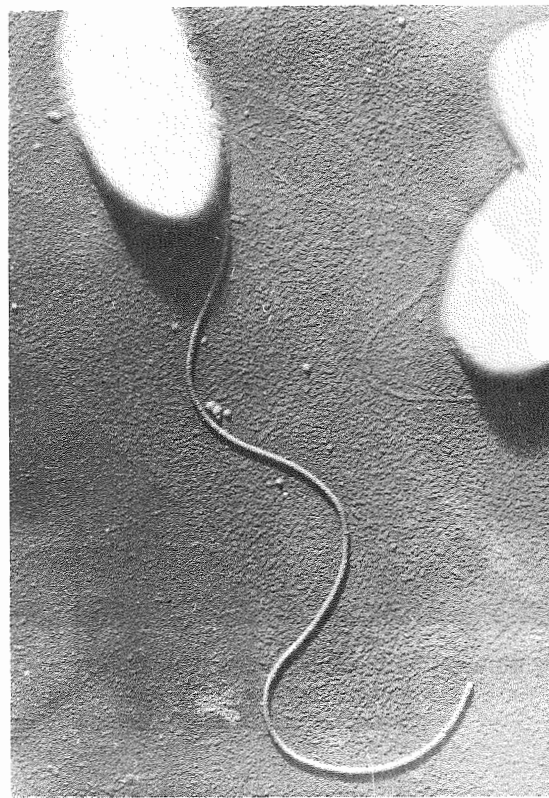


Fig. 30.5 *Pseudomonas aeruginosa*, a species of bacillus with a single flagellum. For this photograph, taken with an electron microscope, the sample was coated with a very thin film of heavy metal, a process known as *metal shadowing*. F. BOVER AND R.C. HAMILTON, COMMONWEALTH SERUM LABORATORIES

plague are the result of such toxins. Sometimes disease symptoms are an allergic response to products of the bacterium. Most of the symptoms of tuberculosis and rheumatic fever result from such allergic reactions.

Viruses

The discovery, by scientists such as Pasteur and Koch in the late nineteenth century, that some diseases were caused by bacteria, was a great step forward for medical science. There were, however, certain diseases for which no bacterial cause could be found. For example, Pasteur tried

in vain to find a bacterium that caused the disease rabies. The cause of these diseases were **viruses**, structures too small to be seen with the ordinary light microscope. When bacterial cultures were filtered, and the filtrate, which was free of bacteria, was injected into a healthy organism, disease symptoms were produced. From this it was realised that there must be a disease-causing agent other than bacteria. Viruses are so small that they can pass through a filter that does not allow the passage of bacteria.

It was not until 1938 that scientists, using an electron microscope, first saw viruses. Subsequent studies showed that they had distinctive structures and differing sizes. All viruses were found to contain genetic material in the form of either DNA or RNA (ribonucleic acid), but they never contained both. The molecule containing this genetic information is surrounded by a coat of protein.

The most distinctive feature of viruses is that, unlike their disease-causing organisms, they cannot be cultured in artificial media in the laboratory. They can only grow inside living cells.

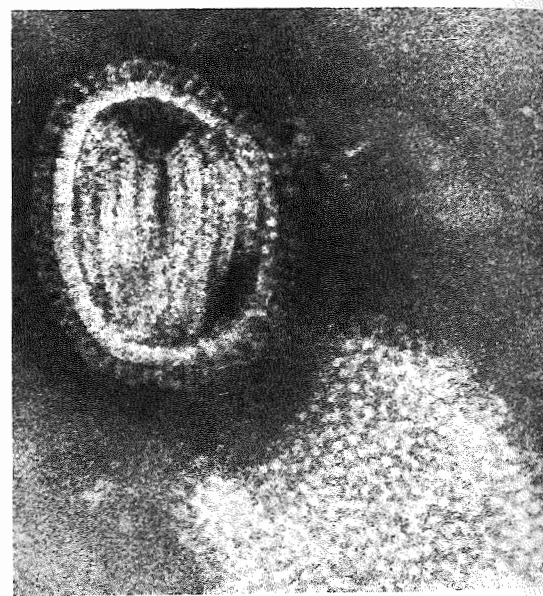


Fig. 30.6 *Influenza virus* ($\times 250\,000$). At the top can be seen the coiled RNA inside a coat with projecting spikes of protein. The particle at the bottom shows the surface with its projecting spikes. J. ARMSTRONG, ROYAL PERTH HOSPITAL

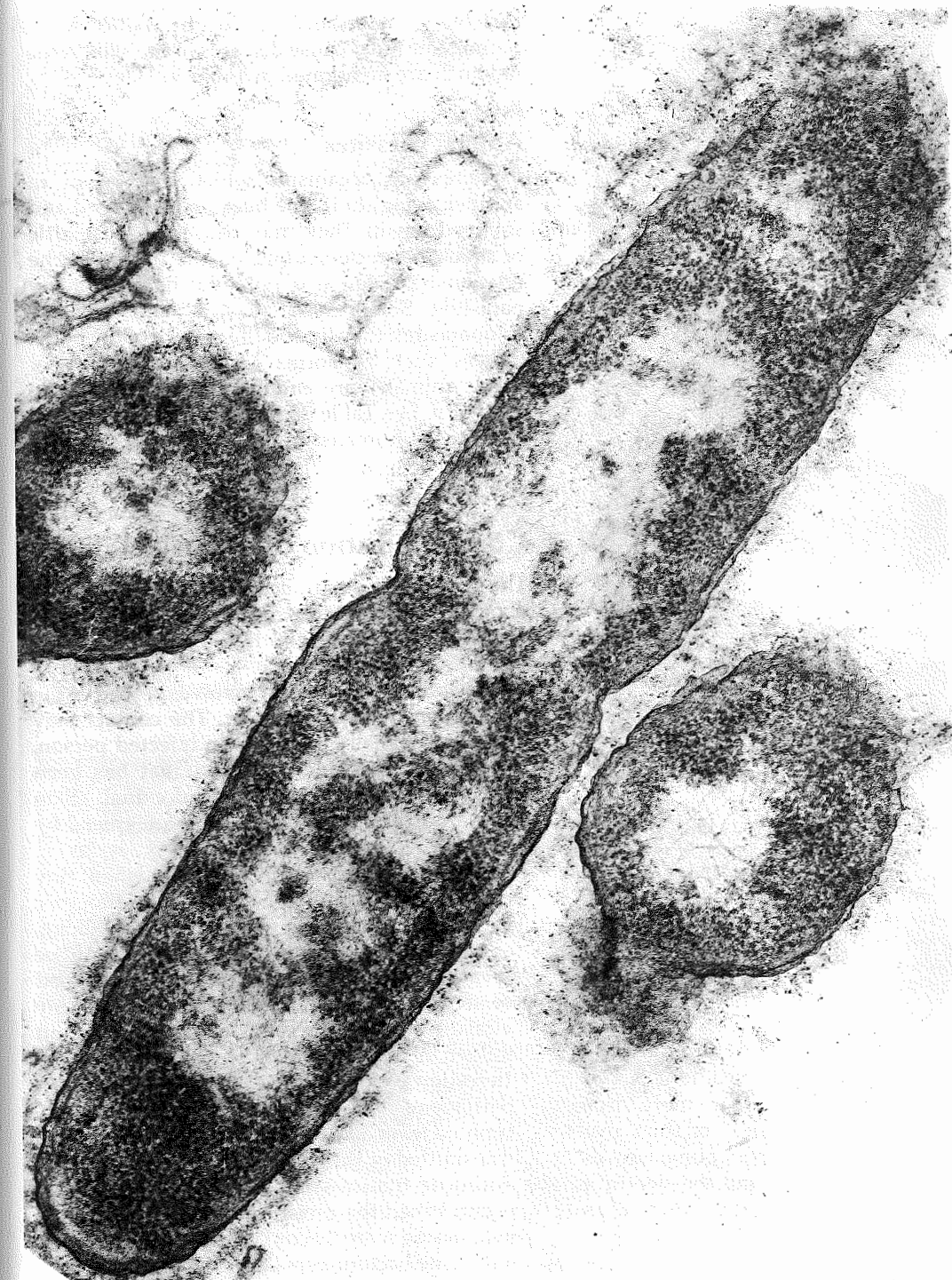


Fig. 30.7 Electron micrograph of *Pseudomonas aeruginosa* ($\times 74\,000$). The elongated structure is a longitudinal section of the bacterial cell. In the middle of the cell is a constriction where cell division is commencing. At each side of the longitudinal section are cross-sections of cells of the same bacterium. Inside all the sections can be seen lighter areas, the nuclear material. The dark lines around the outside of the sections are the cell wall and the cell membrane that forms a multi-layered complex. J. ARMSTRONG, ROYAL PERTH HOSPITAL



Fig. 30.8 *Vaccinia virus* (× 246 000). This is a laboratory cultured cowpox virus used for the production of vaccines against smallpox. It consists of an outer envelope, a biconcave protein coat, which can be seen in the particle at the top, and within that the DNA J. ARMSTRONG, ROYAL PERTH HOSPITAL

When the virus infects a living cell, its DNA or RNA induces the cell to manufacture more virus particles. The new virus particles are then able to leave the host cell to infect others. Some viruses multiply in bacterial cells causing the death of the bacterium. Such viruses are known as **bacteriophages**.

Some diseases caused by viral infections are shown in Table 30.1.

Fungi

Compared with bacteria and viruses, fungi are relatively unimportant in humans. Most pathogenic fungi cause diseases of the skin, some of which are mentioned in Table 30.1.

Animal parasites

Parasites are organisms which live on or in another living thing, the **host**, and gain food and shelter from it. They may cause the host little or much harm depending on the nature of the relationship. Man is a potential host for many parasites. Some, such as fleas and lice, are **ectoparasites**: they live on the surface of the body. Others, such as worms or protozoans (single-celled animals), are **endoparasites**, living inside the body. See Table 30.1 for some of the animal parasites of humans.

Communication of disease

The spread of communicable disease is brought about by the transfer of the pathogenic organism from one person to another. This transfer can occur in a number of ways.

- **Transmission by contact** involves spread of the pathogen by actual contact. The contact may be *direct*, actually touching an infected person, or *indirect*, touching an object that has been touched by an infected individual. Skin infections and venereal diseases are spread by contact.
- **Infection by droplets** may occur when tiny droplets of moisture, harbouring pathogenic organisms, are emitted when breathing, talking, sneezing or coughing. The droplets may be breathed in by others, or may settle on food or utensils to be later ingested with food. Many viral infections, such as those causing colds and influenza, may be spread by droplets.
- **Ingestion** of food or drink contaminated with pathogens may result in disease. Dysentery, typhoid fever and *Salmonella* food poisoning are transmitted in this way.
- **Airborne transmission** of some diseases may occur. When the moisture in exhaled droplets evaporates, many bacteria are killed, but viruses and some bacteria remain viable and can cause infection when inhaled.

Table 30.1 Pathogenic organisms and some of the better known diseases they cause

Bacteria	Viruses	Fungi	Animal parasites
Chlamydia	AIDS	Ringworm	Protozoans
Cholera	Chickenpox	Thrush	Amoebic dysentery
Dental caries (tooth decay)	Cold sores (herpes)	Tinea	Amoebic meningitis
Diphtheria	Colds		Malaria
Gonorrhoea	Encephalitis (viral)		Sleeping sickness
Impetigo (school sores)	Genital herpes		Toxoplasmosis
Legionnaires disease	Glandular fever		
Leprosy	Hepatitis A and B		<i>Platyhelminthes</i> (flatworms)
Meningitis (bacterial)	Influenza		Blood flukes
Plague	Measles		Hydatids
Pneumonia	Meningitis (viral)		Liver flukes
Scarlet fever	Mumps		Tapeworms
Syphilis	Poliomyelitis		
Tetanus	Rabies		<i>Nematodes</i> (roundworms)
Trachoma	Ross River virus		Hookworms
Tuberculosis	Rubella		Roundworms
Typhoid	Shingles		Threadworms
Whooping cough	Smallpox		
	Trachoma		<i>Arthropods</i>
	Warts		Lice
	Yellow fever		Scabies (mite)
			Ticks

- **Transmission by vectors** is the transfer of pathogens by other animals, such as insects, ticks or mites. Some vectors transfer the pathogen directly, as do mosquitoes when

transferring the malarial parasite; others, such as house flies, may spread the pathogen to food or water which is then ingested.

Keeping Australia free of communicable disease

- **Immunisation** Immunisation against a wide range of diseases is available in Australia and Health Departments in all states encourage people to take advantage of the vaccines available. Immunisation will be discussed in more detail in the next chapter.
- **Quarantine** With a few exceptions all people arriving in Australia from other countries are subject to medical inspection on arrival. The main concern is to detect cases of yellow fever, cholera, plague and typhus fever, none of which are endemic to Australia. An **endemic** disease is one that is always present in a country even if the number of cases is very small. Cholera is endemic to South-East Asia and occasionally **epidemics** occur—the number of cases increases to a much higher level than normal. A disease outbreak that spreads from one country to another is said to be **pandemic**. Australian quarantine regulations are aimed at preventing this spread.

Passengers arriving in Australia are required to state their intended place of residence so that, should a case of infectious disease occur among their fellow travellers, they can be traced and treated or isolated if necessary. New arrivals from countries where yellow fever is endemic are required to present a vaccination certificate to show that they have been immunised against the disease.

- **Notification of diseases** In order to control the spread of serious infectious diseases, doctors are required by law to notify state Health Departments should they diagnose certain diseases. These are known as **notifiable diseases**. The Health Acts of Australian states vary but those diseases notifiable in all states in 1988 are listed in Table 30.2. When cases of notifiable disease are reported, health authorities can take early action to prevent the spread of the disease.

Table 30.2 Notifiable diseases in Australia

AIDS	Malaria
Amoebiasis	Ornithosis
Ankylostomiasis	Plague
Anthrax	Poliomyelitis
Arbovirus infection	Q fever
Brucellosis	Salmonella infections
Cholera	Shigella infections
Diphtheria	Smallpox
Gonorrhoea	Syphilis
Hepatitis A	Tetanus
Hepatitis B	Tuberculosis
Hydatid disease	Typhoid fever
Leprosy (Hansen's disease)	Typhus
Leptospirosis	Yellow fever

- **Health regulations** Every state in Australia has a Health Act and many of the regulations in the Act are aimed at preventing outbreaks and spread

of communicable diseases. Disposal of garbage and sewage, handling of food, notification of diseases, purity of water supplies and cleanliness of abattoirs are just a few of the areas governed by these regulations, which help to maintain Australia's good record in the field of communicable disease.

- **Education** Primary schools have, for a long time, taught students basic rules for preventing the spread of communicable diseases. Washing hands before eating, keeping food covered so that vectors such as flies cannot contaminate it, hygienic disposal of garbage and many other basic principles are taught in the classroom. In secondary schools many courses cover health issues. Of particular importance at this level are the sexually transmitted diseases, such as AIDS, syphilis, gonorrhoea and genital herpes. Public education campaigns make use of posters, pamphlets and television advertising to inform the public of disease prevention measures.

Review questions

1. What is communicable disease? Give five examples of such diseases.
2. Explain the difference between:
 - (a) a pathogen and a vector
 - (b) ectoparasites and endoparasites
 - (c) RNA viruses and DNA viruses
 - (d) bacteria and bacteriophages
3. (a) What is the basis for classification of bacteria?
 (b) Describe the main types of bacteria.
 (c) Examine Figure 30.7. How would you classify *Pseudomonas aeruginosa*?
4. (a) Bacteria were first seen in 1683 but viruses were not seen until 1938. Why?
 (b) List four differences between bacteria and viruses.
5. (a) Describe the structure of a bacterial cell.
 (b) Explain how bacteria are able to survive unfavourable environmental conditions.
6. Describe four ways in which communicable diseases are able to be transferred from one person to another.

Apply your knowledge

1. Leprosy (Hansen's disease) is endemic to the Northern Territory of Australia where it is found

1977. In 1966, smallpox, a disease known to have afflicted humans since the beginning of recorded history, affected 1000 million people in thirty-three countries. Eleven years later it was totally eradicated from the earth. Find out how this remarkable medical advance was achieved.

6. The Russian composer Tchaikovsky died of cholera during an epidemic in Moscow in 1893. It is known that Tchaikovsky drank unboiled water

during the epidemic, some believe in a deliberate attempt to commit suicide. Why would drinking unboiled water increase the risk of cholera infection?

7. Hepatitis B is a disease causing growing concern in Australia. Why are medical authorities so concerned about the disease? What precautions can you take to avoid the disease?

CHAPTER 31

Prevention and control of infection

Our bodies have a number of defences that protect us against invasion by pathogenic micro-organisms. Very often we are exposed to pathogens without realising it. Many pathogens are prevented from entering the body, or if they do enter, they are dealt with before they can cause symptoms of disease. Even if we do become ill the defence system often enables recovery to occur without any medical intervention.

Defence against communicable disease may be thought of as homeostasis because it maintains the body at its optimum level of functioning. The body has both external and internal defences against infection by pathogens.

External body defences against disease

The skin provides an effective barrier, which, provided it is not broken by cuts or abrasions, is relatively impervious to the entry of micro-organisms. Sebum, secreted by oil glands in the

skin, contains substances that kill some pathogenic bacteria. Body cavities that open to the exterior are covered by a mucous membrane that inhibits entry of micro-organisms to the organs of the body. The whole of the respiratory, digestive, urinary and reproductive tracts are protected in this way. Hairs, and a layer of mucus in the nasal cavity, enable the nose to trap up to ninety per cent of the particles inhaled when breathing. The trachea (windpipe) and bronchi also have a mucous layer, and the mucus, containing trapped particles and micro-organisms, may be coughed up and swallowed. In the stomach the acidic juices kill many bacteria swallowed in this way or ingested with food. The eyes are protected by the cleansing action of tears, which contain the enzyme **lysozyme**, which kills bacteria. Lysozyme is also found in saliva, sweat, nasal secretions and tissue fluid. Cerumen (ear wax) protects the outer ear against infection. The periodic flow of urine through the urethra has a cleansing action and prevents bacterial growth (Fig. 31.1).

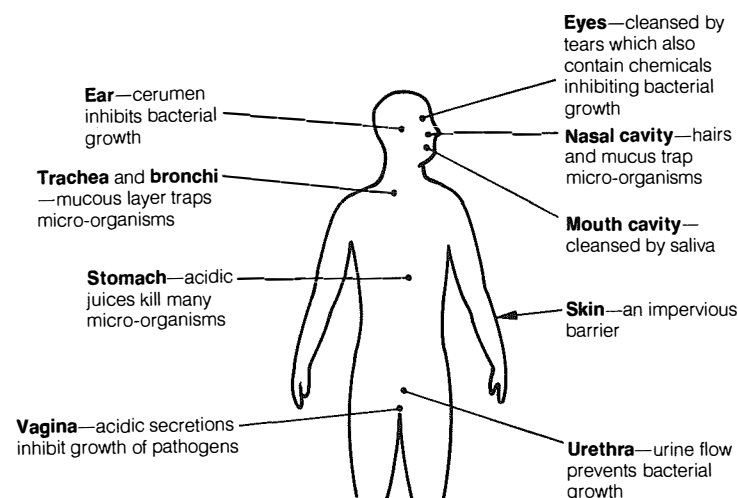


Fig. 31.1 The body's external defences against pathogenic micro-organisms

Internal body defences against disease

If pathogenic micro-organisms should penetrate the body's external defences, there are several internal defence mechanisms that operate to minimise the effect of infection.

The lymphatic system

The lymphatic system consists of:

- a network of lymph capillaries joined to larger lymph vessels;
- lymph nodes, located along the length of some lymph vessels.

The main function of the lymphatic system is to collect some of the fluid that escapes from the blood capillaries and to return it to the circulatory system. In addition to this primary function the lymphatic system is an important part of the body's internal defence against pathogenic organisms.

Lymph vessels

The fluid the lymphatic system returns to the blood is known as **lymph**. Unlike the blood in the circulatory system, the lymph does not circulate—the lymphatic system is a one-way system carrying fluid away from the tissues. The lymph vessels originate as blind-ended tubes in the spaces between the cells of most tissues. These **lymph capillaries** are usually slightly larger than blood capillaries. They are also more permeable than most capillaries. Proteins, and disease-causing organisms in the intercellular fluid can easily pass through the walls of the lymph capillaries into the lymph.

The capillaries join to form larger lymph vessels, also known as **lymphatic vessels**. Lymphatic vessels are similar in structure to veins but they have more valves (Fig. 31.2) and many have lymph nodes along their length. The lymphatics join up and eventually converge into two main channels, which open into veins at the base of the neck (Fig. 31.3). In this way the lymph is returned to the bloodstream.

Lymph is pushed along the lymph capillaries and lymphatic vessels when they are squeezed by the contraction of skeletal muscles or by the pulsing of nearby arteries. The valves ensure that the fluid flows in only one direction. The flow

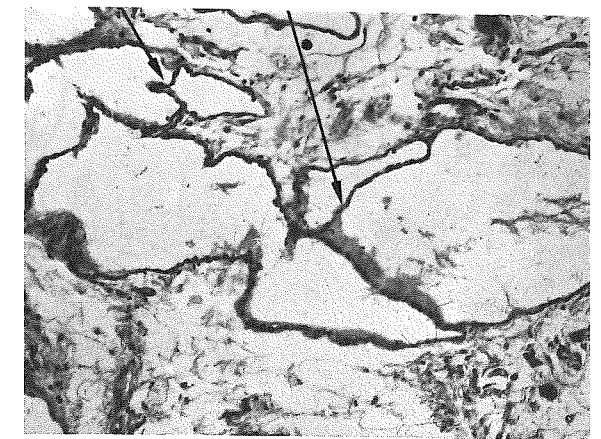


Fig. 31.2 Valves of lymph vessels, indicated by arrows (high power) CENTRE FOR HUMAN BIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

is slow, about 4-5 litres of lymph enters the bloodstream in twenty-four hours.

Lymph nodes

Lymph nodes or **lymph glands** occur at intervals along the lymphatic vessels. They are most numerous in the neck, armpits, groin and around the alimentary canal (Fig. 31.3). Nodes are bean-shaped and range in length from 1 to 25 millimetres. Each is surrounded by a capsule of connective tissue which extends into the node forming a framework. Within the framework are masses of **lymphoid tissue**, containing cells known as **lymphocytes**, **macrophages** and **plasma cells**. Spaces between the lymphoid tissue are criss-crossed by a network of fibres. Lymph enters through vessels on the convex side of the node, filters through the spaces and passes out through vessels on the opposite side (Fig. 31.4). The lymph passes through several nodes before entering the circulatory system.

Role of the lymphatic system in defence against disease

Lymph entering the lymph nodes contains cell debris and foreign particles, and micro-organisms that have penetrated the body's external defences. Some of the micro-organisms may be pathogenic and, if not destroyed, they could cause disease. Larger particles, such as bacteria, are trapped in the meshwork of fibres as the

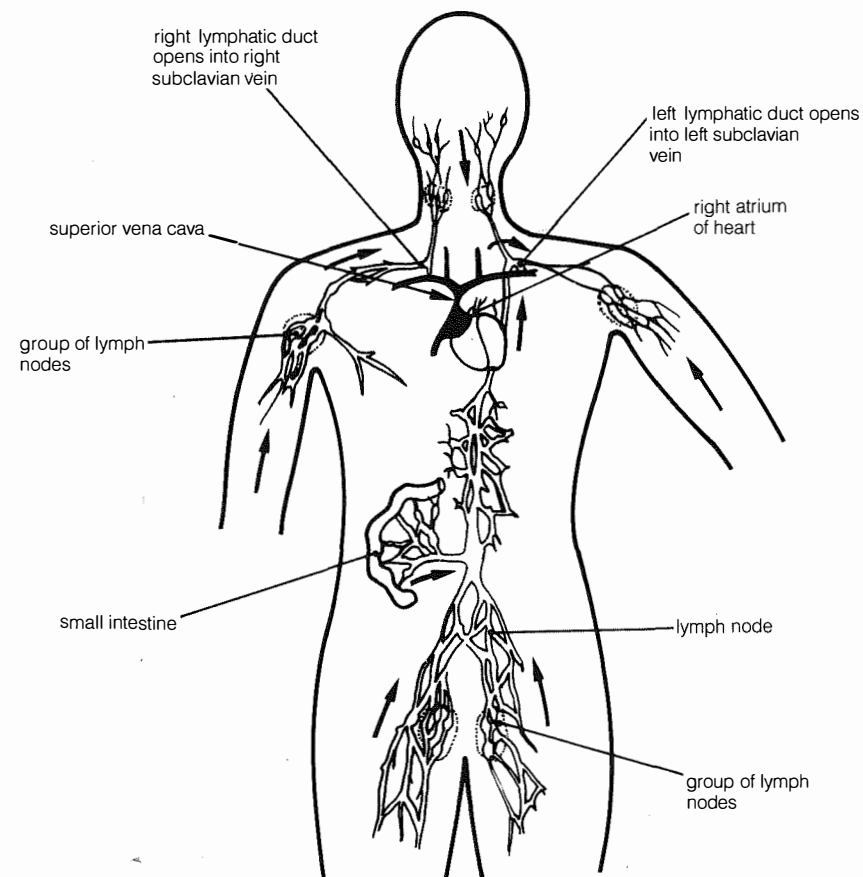


Fig. 31.3 Main vessels of the lymphatic system

lymph flows through the spaces in the nodes. These particles are destroyed by cells called **macrophages**. The macrophages ingest the particles by a process known as **phagocytosis**. Projections from the macrophage surround the particle and take it into the cell where it is destroyed by enzymes (Fig. 31.5). Most bacteria ingested in this way are killed within ten to thirty minutes.

When infections occur, the rate of lymphocyte formation increases and the lymph nodes become swollen and sore. For example, an infected finger may result in swelling and tenderness in the armpit where there are a large number of lymph nodes.

Cells in the lymphoid tissue of the lymph node are involved in the immune response, another mechanism for dealing with infection. This will be discussed later in the chapter.

Lymphoid organs

Several other organs of the body contain lymphoid tissue. These include the bone marrow, the tonsils, the spleen and the thymus gland. Although they contain lymphoid tissue, which is also found in the lymph nodes, these organs have no direct association with the lymphatic system. However, the lymph nodes and the other lymphoid organs are all an integral part of the body's immune system and all produce lymphocytes.

The **tonsils** are small masses of lymphoid tissue embedded in the mucous membrane that lines the pharynx (throat). There are three sets of tonsils. Those at the back of the throat are at their largest during childhood. When enlarged they are called **adenoids**.

The **spleen** lies between the body of the stomach and the diaphragm, towards the rear

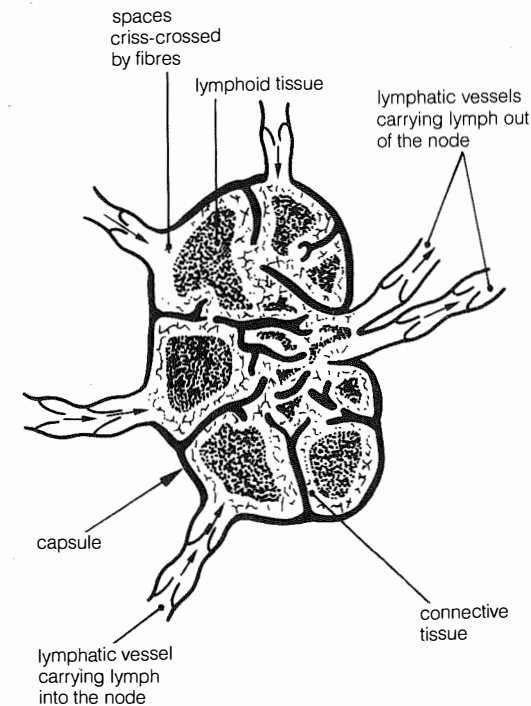


Fig. 31.4 Section through a lymph node. Arrows indicate the direction of lymph flow

of the abdominal cavity. It filters the blood in a way similar to the filtering of lymph by lymph nodes.

The **thymus** is located in the upper chest cavity behind the sternum (breastbone). Its size, relative to the rest of the body, is greatest at birth. At puberty it begins to decline, and in adults it disappears altogether, being replaced by fatty tissue. The important role of the thymus in immunity will be described later.

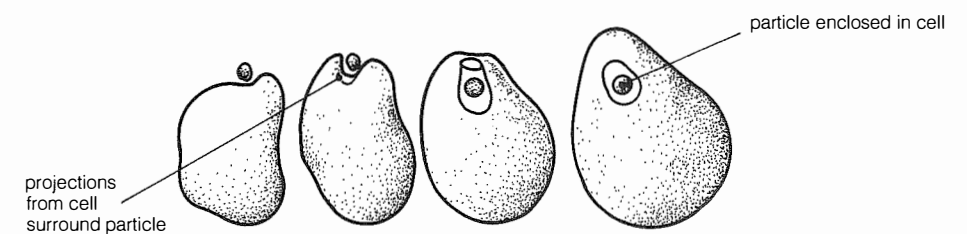


Fig. 31.5 The process of phagocytosis

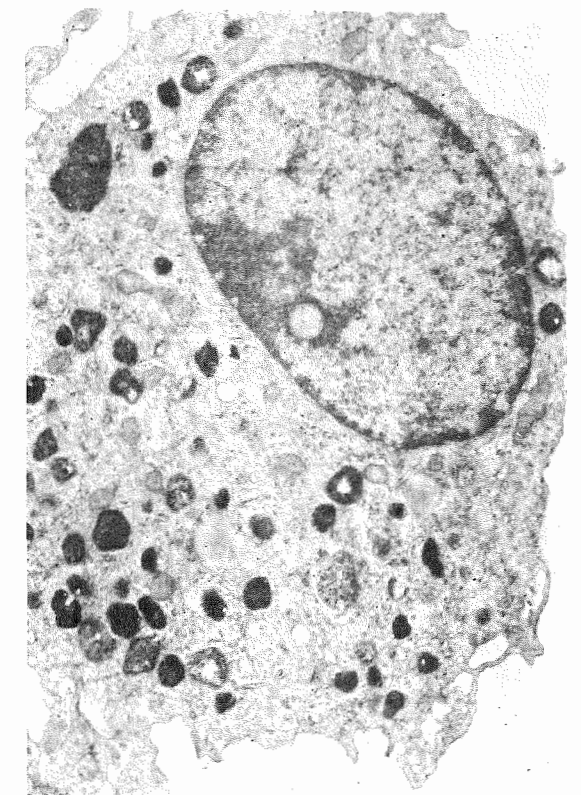


Fig. 31.6 A macrophage ($\times 10\,780$) photographed using an electron microscope. The dark spots are ingested particles. At the bottom right-hand corner are droplets of liquid that have just been engulfed by the cell J. ARMSTRONG, ROYAL PERTH HOSPITAL

Inflammation

Inflammation, or the **inflammatory response**, is a series of events that occurs in response to damage to the tissues. The damage may be caused by heat, chemicals, mechanical injury or infection by micro-organisms. Inflammation can be regarded as a homeostatic mechanism since it acts to neutralise the damaging agent and to return the body to its normal functioning after damage has occurred. It is said to be non-specific because it is a response to any type of tissue damage, no matter what the cause.

The symptoms of inflammation are redness, swelling, heat and pain. Think of a boil, an infected cut, a sore throat resulting from a viral infection or even a pimple. These are all examples of inflammation and all show the four symptoms.

The following events occur during the inflammatory response:

- Small blood vessels in the tissues where damage or infection has occurred dilate. This increases the blood flow, which produces the heat and redness associated with inflammation.
- The permeability of blood vessels in the area increases. This allows fluid and defensive

substances to move out of the circulation and into the inflamed tissue, resulting in the swelling that is one of the symptoms of inflammation.

- White blood cells (**leucocytes**) migrate from the blood into the intercellular spaces around the affected cells.
- Some of the leucocytes becomes macrophages. These and other phagocytic white blood cells engulf and destroy infecting micro-organisms, other foreign material and cell debris.
- **Pus** is produced, unless the inflammation is very mild. It is a thick fluid, which contains dead and living leucocytes and cell debris. If the pus is unable to drain away, the accumulation of pus is called an **abscess**.

A number of chemical compounds are involved in the inflammatory response to tissue injury or infection. One, of which you have probably heard, is histamine. **Histamine** is present in many cells and in blood platelets. It is released from those cells that contain it in response to any damage or infection in the tissues. Histamine causes the dilation and increased permeability of blood vessels associated with inflammation.

The immune response

The **immune response** is a homeostatic mechanism. When micro-organisms or foreign substances enter the body the immune response helps to deal with the invasion and restore the internal environment to its normal condition.

Unlike the defences we have studied so far, the immune response is **specific**. Other defences protect the body against any type of pathogen, but each immune response is directed towards a particular micro-organism or disease-causing substance.

There are two parts to the immune response. One part, called **humoral response** or **antibody-mediated immunity** involves the production of special proteins called antibodies, which circulate around the body and attack invading agents. The second part, called **cell-mediated response**, involves formation of special lymphocytes that destroy invading agents. Both of these aspects of the immune response involve lymphoid tissue. Most **lymphoid tissue** is in the lymph nodes but it also occurs in other parts of the body, such

as the spleen, the thymus gland and the tonsils. Much of the lymphoid tissue is made up of lymphocytes of two types, B cells and T cells. **B cells** provide antibody-mediated immunity and **T cells** provide cell-mediated immunity. Both these cells are produced in the bone marrow, and both end up in the lymphoid tissue, but they mature by following two different routes between bone marrow and lymphoid tissue. About half the cells produced by the bone marrow go to the thymus where they are changed into T cells before being incorporated into the lymphoid tissues. The other half of the cells mature in the bone marrow to become B cells and then also become part of the lymphoid tissues. (T cells are so-called because they mature in the thymus; B cells because they mature in the bone marrow.)

Antigens

Both antibody-mediated and cell-mediated immunity are triggered off by antigens. An **antigen** is any substance capable of causing a specific immune response. Such a substance,

introduced to the body, causes the body to produce specific antibodies. Antigens are large molecules. They may be proteins, carbohydrates, lipids or nucleic acids. An antigen could be a whole micro-organism, such as a virus particle or bacterial cell, or it could be part of a bacterium, such as the flagella, cell wall or capsule. Toxins produced by bacteria are also antigens. Antigens are not necessarily associated with micro-organisms. Tissues transplanted from another person, blood cells of a foreign blood group or things such as pollen grains or egg white contain antigens.

Large molecules produced in a person's own body do not cause an immune response. These are called **self-antigens**. Foreign compounds that do trigger an immune response are **non-self-antigens**.

Antibodies

An **antibody** is a specialised protein that is produced in response to a non-self-antigen. Antibodies belong to a group of proteins known as **immunoglobulins**. The antibody produced in response to an antigen can combine with that antigen to form an antigen-antibody complex. Antigen molecules have specific active sites and at these sites the antibody can combine with the antigen. The active site on the antigen and the active part of the antibody fit together like a key in a lock. Each antibody can only combine with one particular antigen in the same way that a key will only open a particular lock (Fig. 31.7).

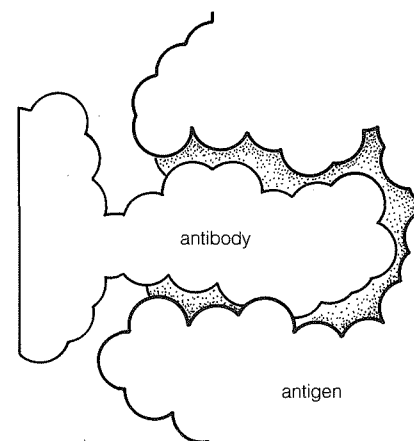


Fig. 31.7 The active sites on the antibody and antigen molecules are complementary: they fit together like a lock and key

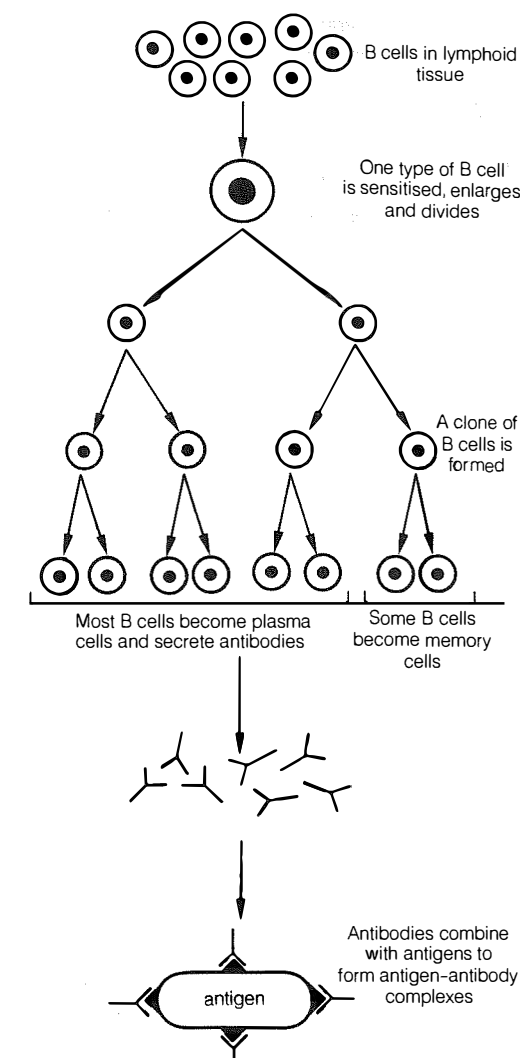


Fig. 31.8 Events in the antibody-mediated immune response

Antibody-mediated immunity

The **humoral response** involves the production and release of antibodies into the blood and lymph. This is antibody-mediated immunity. It provides resistance to viruses, bacteria and bacterial toxins before these micro-organisms or substances enter the body's cells.

Lymphoid tissue contains thousands of different types of B cells. Each type is capable of responding to a specific antigen. When B cells are activated by that antigen they enlarge and divide into a group of cells called a **clone**. Most

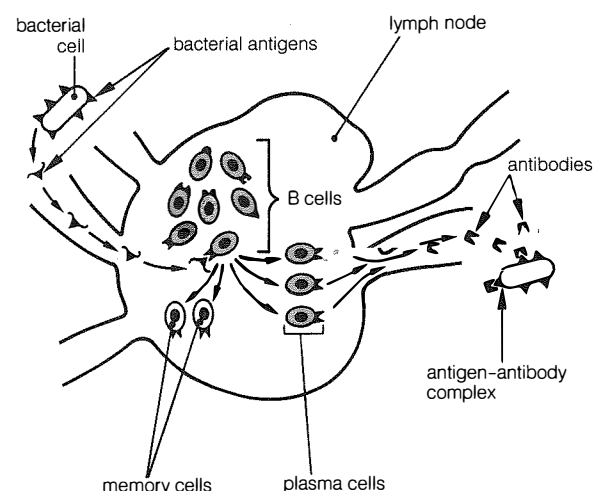


Fig. 31.9 Antibody-mediated immune response in a lymph node

of the clone becomes **plasma cells**, which secrete the specific antibody capable of attaching to the active site of the antigen. These antibodies circulate in the blood, lymph and extracellular fluid to reach the site of the invasion of micro-organisms or foreign material (Figs 31.8 and 31.9). The B cells of the clone that did not differentiate into plasma cells remain as **memory cells**. These memory cells spread to all body tissues to allow the response to occur more rapidly should the antigen enter the body again in the future.

On the first exposure to an antigen the body's immune system usually responds fairly slowly. Several days may be required to build up large amounts of antibodies. With a second or subsequent exposure to the same antigen the response is much faster because of the activity of the memory cells. Figure 31.10 shows the amount of antibody in the blood plasma after a first and second exposure to an antigen.

How do antibodies work to provide resistance to infection? All antibodies combine with the antigen for which they are specific to form an antigen-antibody complex. The response that then occurs varies according to the particular antigen and antibody. Antibodies may:

- combine with foreign enzymes or bacterial toxins or inactivate them by inhibiting reaction with other cells or compounds;
- bind to the surface of viruses and prevent the viruses from entering cells;

- coat bacteria so that the bacteria are more easily consumed by phagocytes;
- cause particles such as bacteria, viruses or foreign blood cells to clump together—a process known as **agglutination** (Fig. 31.11).
- dissolve organisms;
- react with soluble substances to make them insoluble and thus more easily consumed by phagocytes.

Cell-mediated immunity

Cell-mediated immunity (or cellular immunity) provides resistance to the intercellular phase of bacterial and viral infections. It is also important in providing resistance to fungi and parasites and is involved in the rejection of transplants of foreign tissue. Cell-mediated immunity may also be important in fighting cancer cells.

The T lymphocytes are responsible for cellular immunity. They occur in the same lymphoid tissue as the B cells but occupy different areas of the tissue. Like the B cells there are thousands of different types of T cells and each type only responds to a particular antigen. When a foreign antigen, such as a virus or bacterium, enters the body the particular type of T cells, which are specifically programmed for that antigen, become activated or **sensitised**. The sensitised T cells enlarge and divide, each giving rise to a clone, a group of identical T cells (Fig. 31.12). Some cells

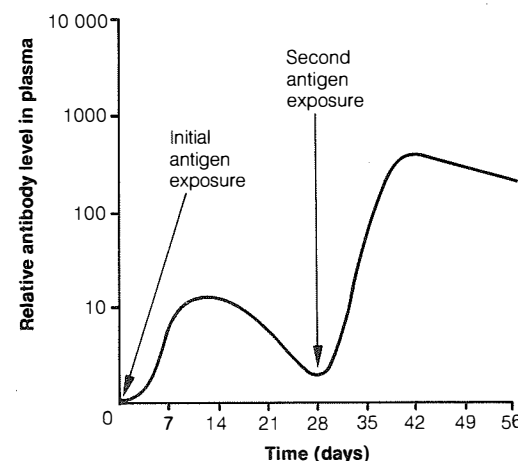


Fig. 31.10 The antibody level in blood plasma after a first and a second exposure to an antigen

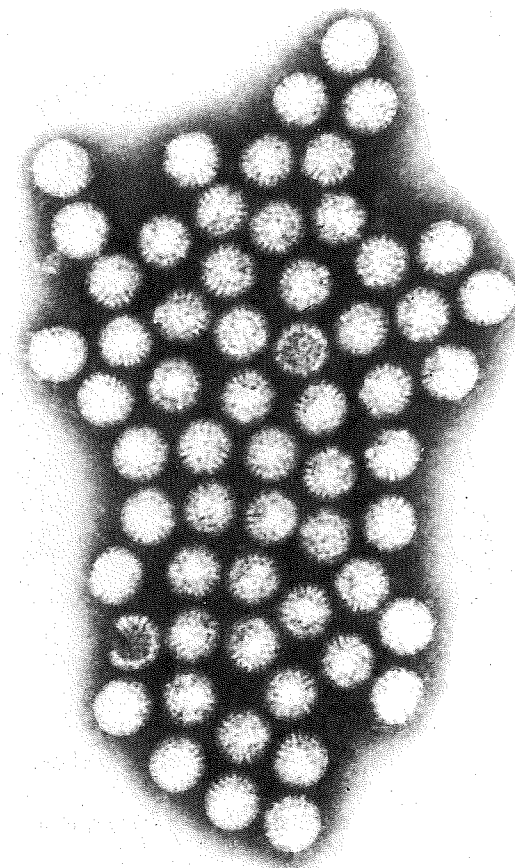


Fig. 31.11 Electron micrograph of virus particles that have been agglutinated by antibody molecules ($\times 116\,000$) I. CHEYNE AND R.C. HAMILTON, COMMONWEALTH SERUM LABORATORIES

of the clone become **killer T cells** and migrate to the site of infection. Other T cells remain in the lymphoid tissue as **memory cells** which are able to recognise the original invading antigen. If infection with the same antigen should occur again these memory cells can initiate a much faster response to the second and subsequent infections.

The killer T cells that migrate to the site of the infection can deal with the invading antigen by:

- attaching themselves to the invading cells and secreting a substance that destroys the antigen;
- secreting a substance that causes lymphocytes at the infection site to become sensitised. The

sensitised lymphocytes then take on the characteristics of sensitised T cells, thus intensifying the response;

- secreting a substance that attracts macrophages to the place of infection so that the macrophages can destroy the antigens by phagocytosis (Fig. 31.13);
- releasing a substance that intensifies the phagocytic activity of macrophages.

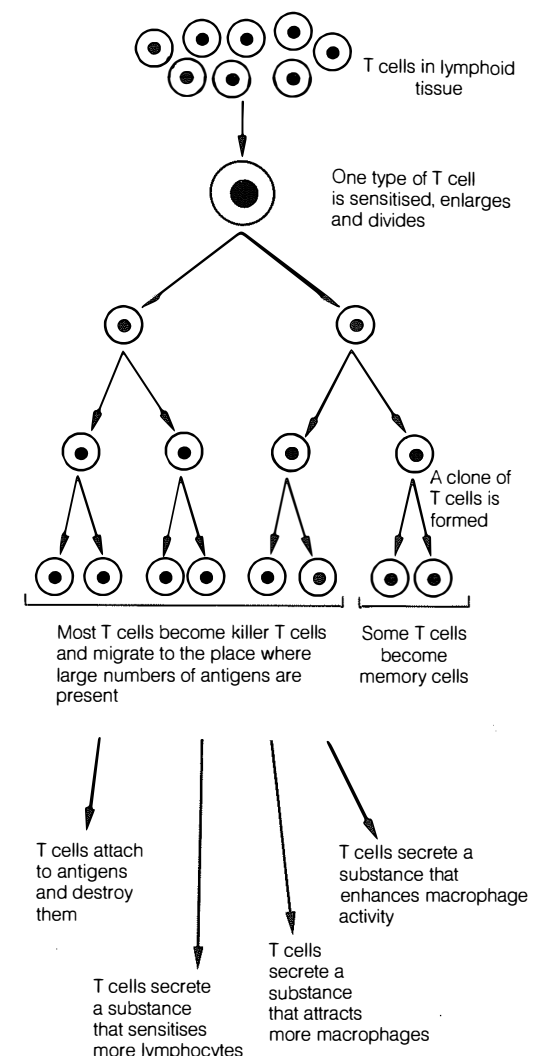


Fig. 31.12 Response to T cells in cell-mediated immunity

reaction are called **allergens**. Common allergens include pollen, dust, antibiotics, certain foods, and chemicals in plants and cosmetics.

On the initial exposure to the allergen a person is said to be sensitised. The antibodies produced during the first contact remain attached to the cells that produce them. On second and later exposures the antigen-antibody reaction destroys the cells as well as the allergens. Destruction of those cells causes release of histamine, a chemical that increases the permeability of blood vessels (see p. 376). As a result of the increased permeability, fluid moves from the blood into the intercellular spaces, causing swelling of the tissue. If the effects of histamine are prolonged, **anaphylactic shock** can occur. This is a constriction of the air passages in the lungs, causing breathing difficulties, and a low blood pressure due to loss of fluid into the tissues. Death can result. Severe allergic reactions, such as those to a bee sting or to penicillin, can result in anaphylactic shock.

If the antigen to which a person is sensitised can be identified, the person can be gradually

desensitised by injections of small but increasing quantities of the antigen that caused the allergic reaction.

Hay fever is an allergic response to proteins in such things as pollen, dust, feathers, fur or some foods. The swelling of tissues due to release of histamine occurs in the nasal passages and in the air passages in the lungs. This causes watery fluid to run from the eyes and nose. **Bronchial asthma** is usually an allergic reaction. The muscles around the small air passages in the lungs constrict making it difficult for air to move in and out. An asthma attack is characterised by wheezing and difficulty in breathing.

There is a tendency for allergies to be inherited. Allergic parents have a higher proportion of allergic children than non-allergic parents. Medical authorities feel that the occurrence of allergies is increasing. However, allergies are nothing new. In about 3000 BC, the Egyptian pharaoh, Menes of Memphis, is said to have died after a sting from a large wasp.

Helping the body overcome disease

Antibiotics

Sometimes the body's immune system is unable to deal with an invading micro-organism quickly enough. The micro-organism multiplies in the body and symptoms of disease can occur. A large number of chemicals that can be used to kill infecting micro-organisms are now available. These are **antibiotics**. Effective antibiotics must be toxic only to the micro-organism and not to the patient's own cells. Each antibiotic attacks a specific target in the micro-organism's cells. For example, penicillin prevents synthesis of bacterial cell walls, tetracyclines and erythromycin affect bacterial ribosomes and nystatin affects the membrane of fungal cells.

Development of antibiotics has had a dramatic impact on human health. Many infections that in the past had devastating effects on society can now be easily treated. Although antibiotics are available for most bacterial and fungal diseases, there are only a few drugs available for the treatment of viral diseases. Development of antiviral medicines is an important area of pharmaceutical research. Another problem with antibiotics is that the bacterium or fungus, through natural selection, gradually develops a

resistance to the drug. For example, penicillin was very successfully used to treat the sexually transmitted disease gonorrhea. In 1975 cases of gonorrhoea that could not be cured by penicillin were first reported. The bacterium that causes the disease had developed a resistant strain. Researchers are continually looking for new antibiotics to replace those to which pathogens have become resistant.

Immunity and vaccines

Immunity is resistance to infection by invading micro-organisms. The body is able to respond quickly enough to prevent any invasion by pathogenic micro-organisms. Such ability to respond rapidly may be natural or it may be artificial. **Natural immunity** occurs without any human intervention; **artificial immunity** results from giving a person an antibody or an antigen that triggers off the immune response so that the person will make their own antibodies.

Natural and artificial immunity can both be either passive or active. **Passive immunity** is when a person is given antibodies produced by someone else. The individual's body plays no part in the production of antibodies. This can occur *naturally* when antibodies from the mother pass across the placenta to a developing foetus or

when the mother's antibodies are passed to the baby in breast milk. It can also be gained *artificially* when a person is injected with antibodies to combat a particular infection. This is often done when a person is exposed to pathogens that cause serious diseases, such as tetanus, diphtheria and rabies. Antibodies are given so that immunity is established immediately. Passive immunity is short lived—it only lasts for as long as the antibodies are in the body.

Active immunity results when the body is exposed to a foreign antigen and manufactures antibodies in response to that antigen. This type of immunity is prolonged because, although the amount of the antibody produced gradually decreases, once the antigen has been dealt with the 'memory' of that antigen persists through the memory cells. Should a subsequent infection involving the same antigen occur, the appropriate antibodies can be produced very quickly before the infection can produce any disease symptoms (Fig. 31.10). Such immunity lasts for many years, often for life. Active immunity to a disease can result from an actual attack of the disease, *natural* active immunity, or from an injection of the antigens associated with the disease, *artificial* active immunity. Table 31.3 summarises the types of immunity.

Table 31.3 Types of immunity

	Natural	Artificial
Passive	Antibodies enter the bloodstream across the placenta or in breast milk	Antibodies are injected into the bloodstream
Active	Ability to manufacture antibodies results from an attack of a disease	Ability to manufacture antibodies results from an injection of an antigen

Immunisation

Persons who have had diseases such as diphtheria, chickenpox, measles, mumps and poliomyelitis are normally immune to these diseases for the rest of their lives. This is due to the natural, active immunity described above. **Immunisation** or **vaccination** is the artificial introduction of antigens of pathogenic organisms so that the ability to produce the appropriate

antibodies is acquired without the person having to suffer the disease.

The **vaccine** is the antigen preparation used in immunisation. Older style vaccines are of three types. One type contains living **attenuated** micro-organisms—micro-organisms of reduced **virulence**; that is, micro-organisms with a reduced ability to produce disease symptoms, so that the immunised person does not contract the disease but does manufacture antibodies against the antigen. Pasteur first used this method to immunise chickens against fowl cholera, and sheep and cattle against anthrax. He attenuated the bacteria by exposing them to a temperature of 42 degrees Celsius (5 degrees higher than normal body temperature) for about a week. Some vaccines containing living attenuated micro-organisms are those for immunisation against polio, tuberculosis, rubella (German measles), measles, mumps and yellow fever.

A second type of vaccine contains dead micro-organisms. Immunity produced in this way is not usually as prolonged as with immunisation using live attenuated micro-organisms. Examples of vaccines of this type include cholera, typhoid and whooping cough vaccines.

The third type of vaccine is made from filtrates of bacterial cultures containing toxins. In cases

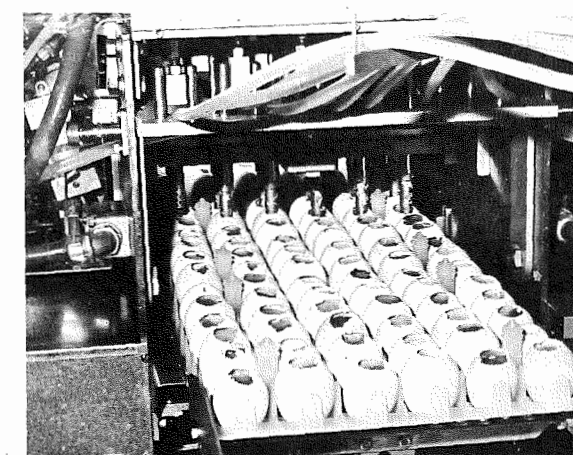


Fig. 31.14 Vaccines for some viral diseases, such as influenza, are produced by allowing the viruses to multiply in fertile eggs. After several days the fluid from the egg is withdrawn, as seen in this photograph. The fluid is then used for preparation of the vaccine V. DRAFFIN, COMMONWEALTH SERUM LABORATORIES

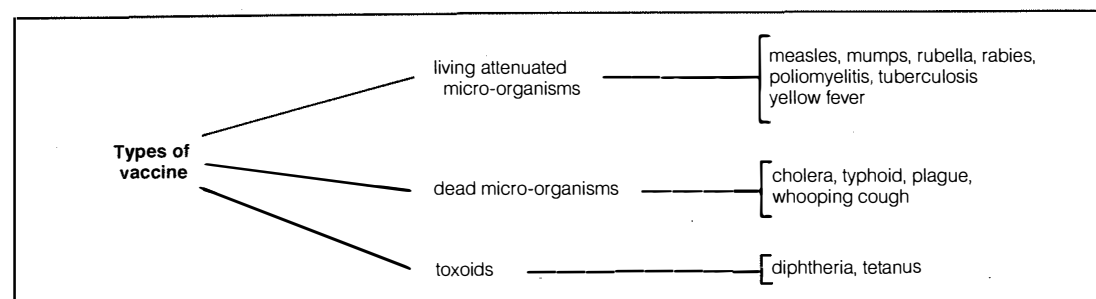


Fig. 31.15 Types of vaccines involved in immunisation against some diseases

where bacteria produce their effects in humans by liberating toxins it is not necessary to use living or dead bacteria for immunisation. The toxins produced by the bacteria can be inactivated so that when injected into a person the person does not become ill. Such inactivated toxins are called **toxoids**. Injections of toxoids are used to immunise against diphtheria and tetanus.

These older types of vaccines have been very effective, but they do have certain problems. With some there is the risk of side effects. For example, about one child in every million vaccinated with dead whooping cough bacteria dies from damage to the nervous system. Some older vaccines, such as the cholera vaccine, give good protection against the disease but are only effective for a short period. Developers of modern vaccines try to produce vaccines that are effective for prolonged periods and produce no side effects. One approach that has been used is to modify the characteristics of the pathogen by slightly changing the DNA in the micro-organism's cell so that the pathogen is less virulent. Another method is to insert certain DNA sequences from the pathogen into harmless bacterial cells. The DNA sequence chosen causes production of antigens that are characteristic of the pathogen. Vaccination with the harmless bacterium results in immunity against the pathogen. It is likely that a great many future vaccines will be made using this **recombinant DNA** method.

Table 31.4 shows a recommended immunisation schedule for Australians from birth through adult life. When travelling outside Australia other vaccinations, such as cholera, yellow fever and typhoid, may be recommended, depending on the country to which one is travelling.

Acquired immune deficiency syndrome

Acquired immune deficiency syndrome, commonly known as **AIDS**, is caused by infection with a virus that causes progressive damage to the body's immune system. As a result of the damage, the patient becomes more susceptible

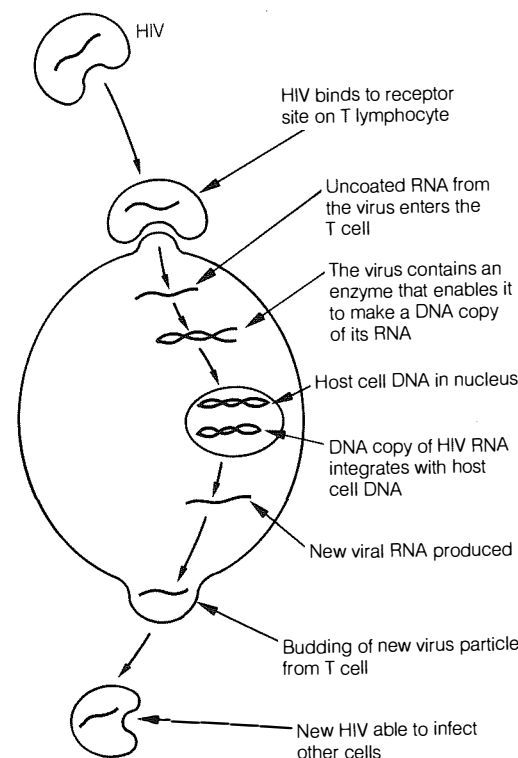


Fig. 31.16 The life cycle of human immunodeficiency virus

Table 31.4 Recommended immunisation schedule for Australians

Age	Disease	Vaccine	Method of introduction to the body
2 months	Diphtheria, tetanus and whooping cough Poliomyelitis	Triple antigen (DTP) Sabin vaccine (oral polio vaccine—OPV)	Injection into the muscle By mouth
4 months	Diphtheria, tetanus and whooping cough Poliomyelitis	Triple antigen (DTP) Sabin vaccine	Injection into the muscle By mouth
6 months	Diphtheria, tetanus and whooping cough Poliomyelitis	Triple antigen (DTP) Sabin vaccine	Injection into the muscle By mouth
12 months	Measles, mumps, rubella	Measles, mumps, rubella vaccine	Injection under the skin
18 months	Diphtheria, tetanus and whooping cough	Triple antigen (DTP)	Injection into the muscle
5 years	Diphtheria and tetanus Poliomyelitis	Child diphtheria and tetanus (CDT) Sabin vaccine	Injection into the muscle By mouth
12–15 years	Diphtheria and tetanus	Adult diphtheria and tetanus (ADT)	Injection into the muscle
Girls 12–14 years	Rubella	Rubella vaccine	Injection under the skin
Adults at 10-year intervals	Diphtheria and tetanus	Adult diphtheria and tetanus (ADT)	Injection into the muscle

to infection and to some forms of cancer. The patient usually dies from a secondary infection or from cancer.

AIDS was first described in the United States in 1981 although it was later realised that an unknown disease recorded in Africa in the late 1970s was, in fact, AIDS. In 1983 the virus that causes the disease was isolated. It is known as **human immunodeficiency virus** or **HIV**. HIV is an RNA virus, consisting of a coat of protein surrounding a core of ribonucleic acid, RNA. It is similar to other viruses in that it is unable to reproduce itself. It must first infect another cell and use that cell's chemical processes to produce new virus particles (Fig. 31.16).

HIV differs from other viruses in several important ways:

- It can infect a cell and remain dormant for a very long period (five years or more). Anyone infected with the virus remains infected for life.
- HIV infects lymphocytes and macrophages, which normally protect us from infections and

from the spread of cancer. Widespread infection of these cells leads to destruction of the immune system.

- The proteins in the outer coat of the virus, which act as antigens, can change from one generation of viruses to the next. This may partly explain why antibodies a person develops against the initial antigen rapidly become ineffective.

With other viral infections, such as colds or influenza, the activity of the body's immune system enables a person to eventually eliminate the virus. Because of the special characteristics of HIV, a person infected with the virus can never eliminate it.

Infection with HIV may be diagnosed by tests that detect the presence of an HIV antibody in the blood. However, as with all infectious diseases, there are not enough antibodies during the early stages of infection. In the case of HIV, this period between infection and the appearance of antibodies in the blood is between two and

who have the virus in their bodies are potentially infectious for life.

AIDS in Australia

The first case of AIDS in Australia was diagnosed in 1982. By mid-1989, almost 1500 cases had been reported in Australia. It has been estimated that for every diagnosis there may be 20 to 30 people who are infected but show no symptoms. Thus, it is likely that there are over 25 000 people in Australia who are infected with the virus, and most of these are currently in good health. The World Health Organization estimated, in October 1988, that world-wide five to ten million people were infected with the virus.

Preventive measures

How does one make sure that there is no risk of becoming infected with HIV? There are several preventive measures you can, and should, take:

- Have sexual intercourse with only one person. There is no risk of AIDS if you only ever have sex with one person *provided* that person never has sex with anyone who could be infected.
- If you do have sexual intercourse with more than one partner, if your partner has sex with others or if your partner shares intravenous needles, *always* use a condom.
- Don't share anything with human blood on it, such as needles or syringes.
- Wash and cover any sores on your skin and don't let sores come into contact with human blood.

There is no risk of getting HIV from such activities as hand washing, hugging, kissing, sharing cutlery or crockery, sneezing, coughing, playing sport or swimming in a pool with others.

Review questions

1. Describe the external defences that prevent entry of pathogenic organisms into the body.
2. (a) What are the components of the lymphatic system?
(b) Draw a diagram of a lymph node and around your diagram write descriptions explaining how the node functions.
(c) Describe the functions of the lymphatic system.

3. List the differences between:
 - (a) blood and lymph
 - (b) blood capillaries and lymph capillaries
 - (c) blood vessels and lymphatic vessels
4. (a) What is a macrophage?
(b) Describe the process of phagocytosis.
(c) What is the importance of phagocytosis in defence against disease?
5. (a) In what parts of the body is lymphoid tissue found?
(b) Why are these tissues known as lymphoid, even though they are not part of the lymphatic system?
6. (a) Why is the immune response said to be a specific response?
(b) What are the two parts to the immune response and what is the main difference between them?
7. (a) What is an antigen?
(b) Explain the difference between self-antigens and non-self-antigens.
8. (a) What is an antibody?
(b) Explain how it is that an antibody is specific to a particular antigen.
9. Describe the events that occur in a humoral immune response.
10. List the ways in which the antigen-antibody complex helps to overcome the effects of invading micro-organisms.
11. Describe the events that occur in a cell-mediated immune response.
12. List the ways in which killer T cells can deal with an invading antigen.
13. Why is it rare to get a disease such as measles or chickenpox more than once?
14. (a) What is an antibiotic?
(b) There are a large number of antibiotics now available. Why is it necessary for researchers to continually look for new antibiotics?
15. What is the difference between:
 - (a) natural and artificial immunity?
 - (b) active and passive immunity?
16. (a) How could passive immunity be gained artificially?
(b) How could active immunity be acquired naturally?

17. (a) What is a vaccine?
(b) Describe three ways in which older types of vaccines are produced.
(c) How are modern vaccines developed?
(d) Explain why it is difficult to prepare vaccines for viral infections.
18. In what ways does the human immunodeficiency virus differ from other viruses that cause disease in humans?
19. (a) How is HIV transmitted from person to person?
(b) What causes death in patients who develop AIDS following HIV infection?

Apply your knowledge

1. Describe the events that would occur if a cut finger became infected.
2. Why is it that people tend to get fewer colds as they get older?
3. Why is it that diseases such as diphtheria and polio have not been completely eliminated from Australia?
4. Few cases of tetanus have been reported in recent years. Why is it still necessary to be immunised against tetanus?
5. Why are researchers looking for better methods of treating cancer interested in new discoveries about the immune system?
6. Typhoid is caused by a bacillus. To positively diagnose typhoid a sample of the patient's blood is taken and mixed with typhoid bacilli. If the bacilli agglutinate (clump together) the patient has typhoid.
 - (a) Why is this a positive diagnosis for the disease?
 - (b) Could the person be suffering from some other disease?

7. When blood transfusions are given, the blood of the donor should be of the same group as the receiver. If the blood groups are not matched, the donor's red blood cells may clump together as they are transfused into the patient. Why would this clumping occur?
8. Haemolytic disease of the newborn may occur if a mother has Rh negative blood and her developing foetus has Rh positive blood. Some of the foetus's blood may leak across the placenta and mix with the mother's blood. The mother will then produce anti-Rh antibodies, which can move back across the placenta and destroy the baby's red cells. The cause of haemolytic disease of the newborn was discovered in 1940. A vital clue that enabled scientists to determine the cause was that first babies were never affected by the condition unless the mother had previously received a transfusion of Rh positive blood.
 - (a) Why would first babies rarely be affected by this condition?
 - (b) Why would the condition be more likely to occur if a mother had previously received a transfusion of Rh positive blood?
9. It has been possible to keep Australia relatively free of diseases such as typhoid, cholera and yellow fever. Why has AIDS been able to spread so rapidly throughout Australia?
10. Because of the severity of AIDS, scientists around the world are actively researching to find ways to prevent, treat and cure the disease. So much effort is being put into AIDS research that new information is published almost daily. Use journal references and newspaper files to find out the latest information available on preventive measures and the treatment of AIDS.